

22-52-Q4

Q ca) HAC + single

cb) DBSCAN + rationale + procedure

Solution (a)

① 0.1414 is smallest

$S_4, S_5 \rightarrow S_{45}$



$$\textcircled{2} d_{S_1, S_{45}} = \min \{ d_{S_1, S_4}, d_{S_1, S_5} \}$$

$$= 1.1045$$

$$d_{S_2, S_{45}} = 1.0247$$

$$d_{S_3, S_{45}} = 0.9695$$

	S_1	S_2	S_3	S_{45}
S_1	0			
S_2	0.1732	0		
S_3	1.6673	1.6401	0	
S_{45}	1.1045	1.0247	0.9695	0

0.1732 is smallest

$S_1, S_2 \rightarrow S_{12}$



$$\textcircled{3} \quad d_{s_{12}, s_3} = \min \{d_{s_1, s_3}, d_{s_2, s_3}\}$$

$$= 1.6401$$

$$d_{s_{12}, s_{45}} = \min \{d_{s_1, s_4}, d_{s_1, s_5}, d_{s_2, s_4}, d_{s_2, s_5}\}$$

$$= 1.0247$$

$$d_{s_3, s_{45}} = \min \{d_{s_3, s_4}, d_{s_3, s_5}\}$$

$$= 0.9695$$

	s_{12}	s_3	s_{45}
s_{12}	0		
s_3	1.6401	0	
s_{45}	1.0247	0.9695	0

0.9695 is smallest

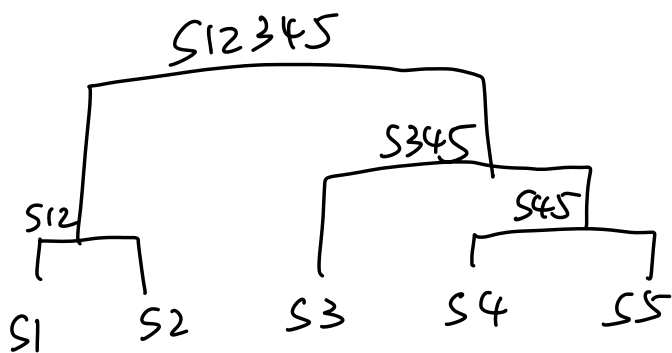
$s_3, s_{45} \rightarrow s_{345}$



$$\textcircled{4} d_{S_{12}, S_{345}} = \min \{ d_{1,3}, d_{1,4}, d_{1,5}, d_{2,3}, d_{2,4}, d_{2,5} \}$$

$$= 1.0247$$

	S_{12}	S_{345}
S_{12}	0	
S_{345}	1.0247	0



(b) ① density - based clustering

eg. DBSCAN

② Rationale

(1) DBSCAN groups points that are closely packed together (high density) and can isolate region of low density as cluster boundaries.

(2) It doesn't require specifying the number of clusters in advance and can handle complex shapes, like nested rings in Figure 2.

③ procedure of DBSCAN

(1) set two parameters:

Min Pts: The minimum number of points clustered together for a region to be considered dense.

eps (ϵ): A distance measure that will be used to locate the

points in the neighborhood of any point.

and randomly pick a data point from the dataset as the starting point.

(2) Find recursively all its density connected point and assign them to the same cluster as the core point

(3) Randomly choose another point among the points that haven't been visited in the previous steps and go to step (2)

(4) This process is finished when all points are visited. Those points that don't belong to any cluster are considered as noise.